

Mainak Pal

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Professional Summary

Computer Science undergraduate with hands-on experience in software engineering, backend development, cloud deployment, and REST API design. Proficient in Java, Python, FastAPI, React.js, REST APIs, AWS, CI/CD, Docker, SQL, and object-oriented programming. Strong foundation in data structures, algorithms, scalable backend systems, authentication/authorization, and cloud-native application development.

Education

B.Tech in Computer Science & Engineering (AIML)

2023–2027

Institute of Engineering & Management, Kolkata

CGPA: 9.10/10 (Till 5th Sem)

Class XII – ISC, Krishnanagar Academy

2023 – 96%

Class X – ICSE, Krishnanagar Academy

2021 – 98.6%

Technical Skills

Languages: Python, Java, C/C++, JavaScript, SQL, Bash

Backend: Flask, FastAPI, Node.js, REST APIs

Frontend: React.js, HTML, CSS

Cloud & DevOps: AWS EC2, AWS S3, GCP, Docker, Git, GitHub, GitHub Actions, CI/CD Workflows, Linux

Database: MongoDB, MySQL

ML/AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Transfer Learning, NLP, CV

Concepts: DSA, OOP, OS, DBMS, REST, JWT, RBAC, Authentication, Authorization, Unit Testing

Experience

Full Stack Developer Intern – StartupCaptain

June–Aug 2026

- Developed RESTful APIs using FastAPI, MongoDB Atlas, and AWS for scalable backend services.
- Implemented JWT authentication, RBAC, email OTP verification, and secure user session management.
- Built responsive React.js frontend integrating cloud-hosted backend APIs using Git-based Agile workflows.

Blockchain Developer Intern – Jalpaiguri Govt. Engineering College

May–Jul 2025

- Built a smart-contract-based access & enforcement system (Solidity, IPFS) with authenticated transaction logging, reducing processing time by **40%**.
- Deployed smart contracts (7.3 TPS, 134ms latency) for scalable performance.

AI Intern – Addauto Technologies

May–Aug 2025

- Built **SENTIVERSE**, a multimodal emotion recognition system using HuggingFace, Whisper, and DeepFace.
- Optimized pipeline via FastAPI; achieved **92% accuracy** and reduced latency by **30%**.

Key Projects

ShikshaSarathi – AI-Powered Accessible Education Platform (Hack4Bihar Finalist)

- Built a full-stack accessibility platform using React.js, Flask, and REST APIs.
- Integrated Whisper and ResNet CNN modules via a modular API layer for ISL detection.

YOUTH0 – Career Recommendation Platform (EY Techathon Finalist)

- Built full-stack recommendation platform using React.js, Flask, MongoDB, and REST APIs.
- Integrated a Random Forest engine into a scalable backend architecture.

Publications

EtherFineChain: A Blockchain-Based Framework for Secure Traffic Violation Logging and Cryptocurrency Penalty Enforcement Using Ethereum Smart Contracts

IEEE ICRCICN 2025

DOI: 10.1109/ICRCICN68210.2025.11364919

Achievements

- Finalist – Smart India Hackathon 2023** (Top 2% of 50,000+ teams), **EY Techathon 5.0** (Innovation Track)
- NPTTEL Elite:** DSA (Top 5%), Introduction to ML
- Certifications:** Cryptography & Information Theory (University of Colorado), Generative AI (IISc Bangalore)
- Solved **140+ LeetCode problems** (DP, Graphs, Algorithms)
- Campus Ambassador:** GirlScript Summer of Code, IIT Kharagpur Kshitij, E-Cell IIT Delhi